

THE HUMAN INCIDENT RESPONSE FRAMEWORK

A Practitioner Wellbeing Framework for Cybersecurity Operations

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This proposed framework adapts validated psychological stress models from the US military, social work, emergency medicine, and child welfare into cybersecurity operational vocabulary. It offers practitioners, peer responders, and team leads a shared severity language, injury taxonomy, exposure mapping, check-in cadence, escalation protocol, and organizational assessment approach for managing the human cost of sustained high-exposure security work.

For the people who keep the systems running and see the incidents nobody else will, and who are expected to come in the next morning as if nothing happened.

For the version of myself who thought being good at the work meant being untouched by it.

And for everyone around us. We built a vocabulary for operational risk and organizational resilience. We never built one for what the work does to the people carrying it. The framework proposed here is an attempt at that.

IF SOMEONE IS IN IMMEDIATE DANGER — CALL 911*

IF SOMEONE IS IN CRISIS — CALL OR TEXT 988*

ANYWHERE IN THE WORLD — [FINDAHELPLINE.COM](https://findahelpline.com)

These are not last resorts. They are first responses. Use them.

* US numbers. Find A Helpline lists free, confidential support lines by country. A fuller set of resources, including practitioner-specific support, is in Part 7, and this box is repeated there. You don't need to finish this document to use any of it.

ABOUT THIS DOCUMENT

Purpose

The aim in publishing this paper and the proposed framework is to make these ideas and proposals available to a wider audience and to inform and encourage discussion, with the ultimate aim of collaboration and development of a refined solution.

This paper was prepared, in part, for presentation at CYBR.SEC.CON on September 16, 2026. The views and frameworks expressed here are subject to revision as the research develops. The author welcomes comments, suggestions, corrections, and challenges.

Not Professional Advice

This paper and the proposed framework are not intended as a substitute for the medical advice of healthcare professionals. They do not constitute professional, legal, or medical advice, and nothing in them should be used to diagnose anyone. Readers should consult appropriate professionals before acting on any information presented here.

Personas and Scenarios

The personas, case studies, and scenarios in this document are fictionalized accounts, constructed for analytical and illustrative purposes from documented clinical presentations in the published literature and from general practitioner experience. They do not depict any specific person, organization, or incident. Any resemblance to an actual person, living or dead, or to a real organization or event, is unintended.

Views of the Author

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THE PREMISE

The cybersecurity industry has a wellness problem and a fluency problem. We are not good at noticing what the work does to our health, and when we do notice, we often do not have the words or expertise to accurately identify the injury.

We have gotten reasonably good at saying the work is hard, demanding, and stressful. We're better about naming the cost to ourselves, our loved ones, and our colleagues. Keynotes and conference talks share personal stories of injuries and acknowledge the practitioner burnout, stress, and toll of the work. None of that is wrong. It just isn't enough. Acknowledging a problem and being able to do something about it are two different things.

What's missing is a shared operational vocabulary for the human side of security work. We already apply that precision everywhere else. There is a rich taxonomy and nomenclature we use to describe a wealth of things in our industry. We have a severity matrix for vulnerabilities. We use standard maturity models for detection capabilities. We have escalation paths, runbooks, and a common language so specific that "Sev2" or "Critical" or "Vulnerability" mean the same thing to everyone in the room. What we don't have is an equivalent for a teammate who's six months into a slow burnout. Or a team lead three weeks past a traumatic exposure that's still replaying on loop. Or an incident responder who is quietly carrying trauma that, technically, wasn't anyone's fault.

The answers exist. They just exist in other fields. Fields that have been attacking and solving this exact problem, under harder conditions, for decades longer than cybersecurity has existed as a profession. Fields that are high-consequence, stressful, and demanding of their practitioners.

The United States Navy and Marine Corps built a four-zone stress continuum to keep war fighters functional under sustained combat exposure. From that work they built Combat and Operational Stress First Aid, a peer-deliverable protocol that doesn't require a clinical license to run, subsequently adapted for fire, EMS, and law enforcement. Emergency medicine organizations built forums for processing the weight of the work as a team instead of alone. Social work and trauma research built the vocabulary and the measurement tools for secondary traumatic stress and vicarious trauma, which is harm that arrives indirectly through exposure to other people's trauma. Child welfare went a step further. They built an instrument that scores the organization, not just the person, on the theory that no amount of individual resilience training fixes a structurally broken system.

None of these fields look like cybersecurity, and yet all of them attacked and answered a version of the same problem: how do you protect yourself and others when the job exposes you, again and again, as a condition of employment, to things most people never have to carry.

The Human Incident Response Framework is an attempt to take a step forward for our industry; to take what these other fields have already painstakingly validated, and translate it into cyber's own language. Not because cyber is combat, or firefighting, or emergency medicine, or social work. It very much isn't, and the differences do matter. But we can still harness a wealth

of published, peer-reviewed injury and stress trauma knowledge, to infer, analyze, and deploy it to help others cope, process, and heal. It might even save lives.

BEFORE YOU USE THIS FRAMEWORK: THE TRAINING FOUNDATION

Read this document first, then go get trained. This framework gives you vocabulary, a triage model, and a protocol for recognizing and responding to stress injury. What it can't give you is the judgment that comes from training, and showing up without that isn't neutral.

Two bodies of work sit underneath everything here. They're structured differently from each other, and it's worth knowing how before you go looking.

Stress First Aid

Stress First Aid came out of the US Navy and Marine Corps, built specifically because the chain of command isn't always the right first contact for someone struggling with the weight of the work. Stress First Aid for Health Care Workers, from the National Center for PTSD, is the same model adapted for a civilian high-stress, high-consequence workforce. Same seven C's, same logic, and it's the version to start with.

It isn't a certification, and there's no central body issuing one. It's a published model, and the manuals are free from the National Center for PTSD. Read them. Stress First Aid gives a non-clinical peer a structured set of actions, the seven C's, that can be delivered without a license, a clinical background, or a formal role. The check-in and escalation guidance in this framework is built on it (see Part 4), and reading the manual before you need it is the difference between running a protocol and fumbling through a conversation hoping something lands. The link is in the resources section.

Instruction does exist if you want it, delivered by various institutions rather than by one authority. Duke University's School of Nursing offers Stress First Aid as a short course with continuing education credit, and a self-paced version through Coursera. The Schwartz Center for Compassionate Healthcare runs courses developed with the model's authors. Adapted versions exist for law enforcement, firefighters, and wildland fire crews.

Mental Health First Aid

Mental Health First Aid works the other way. It's a formal eight-hour certification course, delivered by licensed providers, that teaches you to identify, understand, and respond to signs of mental health and substance use challenges in others. It was developed in Australia in 2001 and is now a global program running in 29 countries. Over 8 million people worldwide have completed the training.

Where Stress First Aid gives you the peer response protocol, Mental Health First Aid gives you the broader clinical literacy to understand what you're looking at before the protocol starts.

Together they cover the two gaps that most well-meaning peers run into: not knowing what to do, and not knowing what they're seeing.

The course is available in person, online, and in hybrid format, and it's free in many jurisdictions. A version exists specifically for public safety professionals. If you're outside the United States, find your country's licensed program at mhfainternational.org.

PART 1 — THE ZONES

A stress injury behaves like an incident. It has indicators. It has a severity. It has a triage and escalation path, and there is a real difference between "this resolves on its own" and "this needs intervention now." Treat it that way and it stops being a vague gesture toward self-care, or an empty platitude for mental wellbeing. It becomes something a team lead can actually run, the same way they'd run any other incident response playbook.

The zones, adapted from the US Navy/Marine Corps Stress Continuum (MCTP 3-30E), are the severity scale for those injuries, one shared vocabulary so that "I'm in the orange this week" means something specific to everyone in the room.

The Zones

Zone	State	Looks Like	What You Do	Owner
● Green	Ready	Steady, sleeping, connected, recovering between incidents	Protect recovery. Rotate high-exposure duty. Normalize the conversation now, before it's needed.	Self + team
● Yellow	Reacting	Short fuse, off sleep, cynicism, pulling back. Normal during a surge.	Stress First Aid: move the load, force the recovery day, restore connection, address the stressor.	Trained peer
● Orange	Injured	Persistent sleep loss, intrusive replay, guilt, numbness, increased substance use, slipping on things that used to be automatic	Cut exposure now. Structured peer check. Warm handoff to care. Follow up until baseline is restored.	Peer + EAP*
● Red	Crisis	Weeks of worsening symptoms, real functional impairment, hopelessness	Clinical escalation. The peer's job is to bridge, not treat.	Licensed professional

* Peer + Employee Assistance Program (EAP) support in the Orange zone depends on the EAP actually having trauma-informed capacity. A generalist EAP is not automatically equipped for secondary traumatic stress, vicarious trauma, or moral injury (see Part 5).

Green is the person who finished the engagement, slept, and came back Monday actually ready. They are fully present, not just functional. You know who this is on your team, because they are engaged, vitalized, and ready to work with the team.

Yellow is your best analyst three weeks into a major incident. They're short with people they're normally patient with, skipping lunch, sending messages at midnight. This is expected, and it resolves when the surge ends. Almost everyone reading this has been here.

Orange is that same analyst six weeks after the incident closed, and they still aren't back. Their sleep is still off, and they're still edgy. The thing that was supposed to be temporary isn't temporary anymore.

Red is when it has been that way long enough, and gotten heavy enough, that a peer's job changes. You're still checking in, but you are now also bridging it to a professional qualified to help someone in a crisis.

The Diagnostic That Matters Most

The most important line in this scale is not the red zone. It's the boundary between yellow and orange.

Did it resolve when the pressure came off?

Yellow is tied to a stressor. When the surge ends, when the engagement closes, when the pressure comes off, yellow should resolve. It is nearly universal, and it doesn't require professional intervention. Everyone who has worked a major incident has been in yellow.

Orange is when the stressor is gone but the person still isn't recovering. That single question is the most teachable, most defensible read a manager or peer can make without a clinical license. It doesn't ask anyone to diagnose. It asks them to notice whether someone bounced back. If the answer is no, the zone is orange, and the response changes.

The Burnout Misconception: Four Injuries, One Zone

The industry has one word for what high-exposure work does to people: burnout. The compression into a single word is understandable. It's an easy word to reach for, and it's often the wrong one. In reality, there are four distinct, clinically and academically validated injuries here, each with a different cause and a different treatment, and calling all of them burnout is how people spend six months on the wrong intervention and wonder why nothing is working.

Here are the four injuries this framework asserts this profession sustains, plainly defined. **These are common presentations, not complete checklists.**

Burnout is the exhaustion that comes from chronic workload without enough recovery. It builds slowly, under sustained operational tempo, and it shows up as depletion, cynicism, and a sense that the work no longer matters. It is a response to load, not to any single event, which is why it eases when the load eases. If someone recovers once the pressure comes off, that was burnout.

Secondary traumatic stress is what absorbing someone else's trauma does to you, and it doesn't much care how the dose arrives. One high-dose exposure can do it. So can hundreds of smaller ones over years. Either way it looks like PTSD: intrusion, avoidance, hypervigilance. What makes it secondary traumatic stress rather than vicarious trauma isn't the timeline, it's that the symptoms look like trauma rather than a changed worldview.

Vicarious trauma is the gradual reshaping of how you see the world after years of exposure to other people's worst days. It is cumulative and slow, and it changes your beliefs rather than your mood: about safety, about people, about how much danger is out there. The person

carrying it usually cannot see the drift, because it happened one reasonable step at a time and there is no longer a baseline to compare against. It is an injury to your worldview, not your workload.

Moral injury is the damage done by a violation of your own values, whether you were forced to act against them, prevented from doing what you believed was right, or made to watch it happen. It presents as guilt, shame, and a loss of meaning, and it is not fatigue. Rest does not touch it, because the wound is to conscience, not to energy. It comes from a betrayal, often by an institution you trusted to do the right thing.

Injury	Trajectory	Dominant Tell	What Catches It	What Treats It
Burnout	Slow depletion under sustained load	Exhaustion, cynicism, inefficacy	Did it recover when the load eased?	Fix the system: workload, recovery time, role clarity
Secondary Traumatic Stress (STS)	Fast, or slow. One high-dose exposure can move you, and so can hundreds of smaller ones.	Intrusion, avoidance, hypervigilance, whether it arrived at once or accumulated	Re-experiencing, whether or not there's one event to point at	Trauma-informed care, dose management
Vicarious Trauma	Years-long drift; person re-baselines and loses the comparison point	Changed worldview, mistaken for "just being realistic"	Longitudinal measurement against old baseline or peers	Deliberate stewardship, clinical work
Moral Injury	Event-triggered, can sit latent for months	Guilt, shame, betrayal, loss of meaning. Not fatigue.	Surfacing the moral weight specifically, not the exhaustion	Moral repair: acknowledgment, accountability, meaning-making

What the Road to Injury Looks Like

Here are vignettes of four practitioners, all in the same orange zone, and they took four different roads to get there. Each one is a fictionalized illustration drawn from documented clinical presentations and practitioner experience across the cybersecurity industry. **They are teaching cases, not diagnostic instruments.**

Dana — DFIR Lead — Burnout

Dana leads digital forensics and incident response (DFIR) for a firm. When a customer declares an incident, she is the one who picks it up, scopes it, and runs it to close as the incident commander. Over one fourteen-day on-call rotation, three separate customers declared at once, and the tempo never let up long enough for her to catch her breath between them.

● **Green:** steady, reliable, the person the team anchors to.

- **Yellow:** three major engagements in fourteen days on call, operational tempo that never relented. Short with people she's normally patient with. Running on fumes but still running.
- **Orange:** the last engagement closed, her on-call shift finally ended, and the load came off. Dana didn't.

She's not burned out because she can't handle the work. She's not burned out because she isn't good at it. She's burned out because she has been drinking from a fire hose with no meaningful break and no lower dose, and at some point the body and the mind stop being able to tell the difference between a crisis and a Tuesday.

What got missed: Nobody was comparing Dana to Dana six months ago. They were comparing her to yesterday's version, and yesterday's version looked fine.

The fix: What Dana needs isn't a long weekend. It's a change to the structure that produced the load: the rotation, the pileup of concurrent engagements, the absence of any real gap to come back to baseline. The **tempo** and the **load** are the variables, not her capacity to do the work well.

Marcus — CTI Analyst — Secondary Traumatic Stress

Marcus was doing routine cyber threat intelligence (CTI) work, monitoring a ransomware cartel's disclosure site. Standard stuff. Except that day the threat actor had hit a pediatric oncology center, and what they published wasn't just the organization's data. It was patient records. Children's names. Diagnoses. Dates of birth. Home addresses. Kids with cancer, and their information is now sitting on a ransomware cartel's shame site for the world to see.

Marcus documented it, filed it, and moved on, because that's the job.

Except he didn't move on.

- **Green:** before that session. Steady, good at the work, no reason to think this day was different from any other.

- **Yellow:** inside it. Nowhere to go with what he was looking at, and the queue didn't stop to let him.

- **Orange:** within days. Replaying what he saw during unrelated work. Avoiding his queue in the mornings. Not sleeping right. Snapping at people he's normally patient with. He told his team lead he'd been feeling off. His team lead told him to take a long weekend.

What got missed: The long weekend isn't the problem. What Marcus experienced in that session left a mark that rest doesn't touch. His manager saw someone who needed a break. What Marcus actually needed was someone to recognize that a specific thing happened, that it was serious, and that it required more than time off.

The fix: Not a long weekend. Dose management, immediate reduction of re-exposure, and a conversation that names what he actually saw rather than routing around it.

Cole — Senior Threat Intelligence Analyst / DFIR Responder — Vicarious Trauma

Cole has been doing cybersecurity in various forms for eleven years. He is good at it. There is no single incident anyone could point to. By every visible measure he is fine. As an incident responder over those eleven years, he has been the calm voice on the phone while someone was in crisis about being a victim of a cyber attack. As a cyber threat intelligence analyst, he has worked global disasters and terrorist attacks and had to research threat intelligence on the dark web that led him to things he cannot unsee. He has carried what he knew about those crises internally, because that is the job. You don't get to fall apart. You are the person who holds it together while someone else falls apart around you. The accumulation of that, over time, is what nobody warned him about.

● **Green:** year one. Engaged, curious, genuinely energized by the work.

● **Yellow:** no single moment. No flagged incident. Nothing he could point to if you asked him when things changed.

● **Orange:** now, and for longer than he realizes. His threat model for everyday life runs the same parameters as his professional one. His neighborhood. His kids' school. His family's devices. He calls it being realistic. People who knew him in year two would not recognize it as realistic. His wife has stopped bringing up certain topics because the conversation always ends the same way.

He cannot see the drift because he has no comparison point left. Each step felt like the natural accumulation of experience. Getting better at reading the world. Losing naivety. Growing up professionally.

What got missed: There is no moment anyone could have flagged. VT evades single-point detection by design. Catching it requires comparing Cole now to Cole at year two, and nobody kept that record.

The fix: Not a check-in. A check-in compares now to recently, which returns normal. What Cole needs is longitudinal measurement, deliberate time in contexts that have nothing to do with threat, and clinical work focused on the worldview itself, not the symptoms sitting on top of it.

Devin — Senior Vulnerability Researcher and Team Lead — Moral Injury

Devin leads a small vulnerability research team. Over four months they found and chained three flaws in a widely deployed product into a working pre-authentication remote code execution. The write-up was clean. The severity scoring was defensible and his team checked each other's work twice before it went to the vendor.

What came back was pressure. The vendor disputed the exploitability, then disputed the scoring, then raised the relationship. By the time the advisory published, the CVSS had been negotiated down, the chained impact was described as theoretical, and the proof of

concept was gone. The advisory went out with Devin's team named as the finders, under a severity none of them agreed with.

- **Green:** leading the team, trusted by them, doing research he believes in. The kind of lead who takes the weight of the call so his researchers don't have to.

- **Yellow:** processing it the way the work sometimes goes. Uncomfortable, but moving. He told himself disclosure is a negotiation and always has been. He told himself the team understood. He expected the discomfort to resolve once the next cycle started.

- **Orange:** it didn't resolve. A question that won't close, running in the background of every subsequent engagement. What is the research actually for if the findings are adjustable for reasons that have nothing to do with the evidence? What does he tell his team when they ask why the score changed? What does he owe them as the person who led the work, as the person whose name is on a document that understates what they found, and what does he owe whoever runs into that flaw next?

His output is normal. He shows up, leads the team, does the work. No check-in has flagged anything outside normal range. The drift from yellow to orange produced no observable signature, because moral injury in the early stages doesn't present as impairment. It presents as someone who has gotten a little quieter.

What got missed: The standard check-in vocabulary of workload, sleep, and how are you holding up doesn't surface a values violation. The right question is more specific: what did that call cost you, and has anyone named that cost out loud. Nobody asked Devin that question. Nobody asked his team either.

The fix: Not rest, not a lighter queue. Acknowledgment of the specific weight, space to name what was compromised, and moral repair that doesn't depend on relitigating the decision or reassigning blame. Devin knows the decision wasn't his to make. That's not the point. The point is that he and his team did the work right, the work got changed, and nobody has acknowledged what that cost.

The Pattern

Dana, Marcus, Cole, and Devin are all in the orange zone. A zone-only read gets you this far: something is wrong, the normal recovery mechanisms aren't working, this needs attention. That's necessary, but it's not sufficient. Dana needs her organization to fix the conditions that depleted her. Marcus needs the exposure stopped and trauma-informed support. Cole needs longitudinal measurement and worldview-level clinical work. Devin needs the moral weight named, not managed around, before anything else will touch it. **The zones get you to the conversation, but the injury type tells you what the conversation is for.**

Nobody Comes to Work Empty

Somebody on your team is doing elder care. Somebody is a single parent. Somebody buried a parent last spring and hasn't mentioned it since. That isn't unusual and it isn't weakness. It's just life, and it runs concurrent with the job.

Go back to the four people you just met. What if Dana was managing her mother's dementia through that same fourteen days? What if Cole was eighteen months out from a divorce? None of it changes the injury, or the fix. What it changes is how much room they had before the work started drawing down, and how fast they moved once it did. It runs both directions too, and the compounding is what gets missed, because each half looks survivable on its own.

The zones measure state, not source. You don't need to know what else someone is carrying, and it isn't your job to find out. What this framework asks is that you don't assume the job is the only variable. If someone is moving through the zones faster than usual, respond to that. The reason can stay theirs.

What This Framework Can't Do

Everything in this Part is a detection aid. It isn't a safety system, and it shouldn't be treated as one.

People can be in serious trouble with none of these indicators present. Someone can hold a normal conversation, hit every deadline, answer a check-in honestly by their own reckoning, and still be nowhere near all right. Someone can also decline to tell you anything, and that's their right.

Which means running this well and still losing someone is possible. If that happens, the protocol wasn't the thing that failed, because the protocol was never the thing holding the weight. Peer support sits alongside clinical care, crisis services, and whatever your organization already has. It replaces none of them, and it doesn't turn a colleague into a safety net.

One practical consequence. If you are worried about someone *right now*, don't run a protocol. Use the crisis resources at the front of this document, and stay with them.

PART 2 — THE EXPOSURE MAP

Not every seat in a security organization carries the same risk profile. Burnout can happen anywhere chronic load exists, but secondary traumatic stress (STS), vicarious trauma (VT), and moral injury are not evenly distributed across a team. They cluster around specific functions, specific content types, and specific decision responsibilities. Treating the org as a uniform risk surface misses the point, and it's how organizations end up applying general wellness resources to roles that need something more specific. The exposure map is triage by function, not title.

What Drives Exposure

Content type: What the person is actually looking at. Explicit material, victim data, human suffering, evidence of atrocity. This is the primary driver of STS and vicarious trauma risk. The content doesn't have to be graphic in a traditional sense. Patient records belonging to children with cancer carry the same load as imagery does, for different reasons.

Decision weight: The degree to which the person's professional judgment has real-world consequences for people outside the organization. When a call can be wrong in ways that cost lives or compromise national security, the conditions for moral injury are structurally present.

Institutional friction: The degree to which the person's professional judgment is subject to override for non-technical reasons. Business risk, legal exposure, client relationships, market considerations. This is the variable that converts a hard job into a values violation.

Cumulative dose: How long and how frequently, independent of severity. A practitioner running daily collection on moderate-severity content for five years carries a different risk profile than one with a single acute exposure. Dose is the multiplier on the other three drivers. It doesn't replace content type, decision weight, or institutional friction, it amplifies all of them over time. Think of it as the difference between a single high-severity alert and a sustained elevated baseline that never returns to green. The alert is STS. The baseline that never drops is where VT and burnout live.

Adversarial engagement: Direct, real-time interface with a hostile actor who is actively causing harm. This is the exposure driver that has no parallel in the fields this framework borrows from. Social workers process trauma after it has occurred. The CTI analyst or threat hunter inside an adversary's infrastructure, or the DFIR responder who finds the adversary still hands on keyboard in an active incident, is watching harm being generated by a conscious actor making deliberate choices, and carrying the weight of that observation while waiting for the operational moment to act. It is also not passive. The analyst and the responder are not only witnesses; they are contending with a live, adaptive opponent for control, or to stop something bad before it happens. That adds a particular kind of pressure: the pressure of not making a mistake against someone who is actively working to make you fail, on a job that is already hard. When the thing being fought over matters enough, the stakes can become stratospheric. The psychological cost sits between STS and moral injury without being fully either. Over a career, that accumulates in ways the field hasn't measured yet and doesn't have adequate words for.

This framework names it as an exposure driver because someone needs to, and because the practitioners who carry it deserve to have it acknowledged.

Secrecy: The degree to which the work can't be discussed outside a defined boundary. A classification, an embargo, a disclosure window, an NDA, an open law enforcement matter. This one doesn't produce an injury of its own. What it does is restrict and inhibit the support that would otherwise reach someone in yellow, or even orange. Supervisory support, coworkers, and people outside work are the best-evidenced protective factors in this literature, and a wall around the work cuts off all three at once. The 2022 RAND study of the intelligence community found that personnel bound by law to secrecy face a heightened risk of trauma and reduced recourse for managing it (Sudkamp et al., 2022). Read it as a condition that raises the severity of whatever else is present, and makes it harder to see.

A Note on the Research Base

The role-to-injury mapping that follows is an inference-based assessment, and it's worth being precise about which parts of it rest on published research and which don't.

Burnout in this profession has been studied directly. There's a body of cybersecurity-specific work going back a decade covering security operations centers, incident responders, and chief information security officers, using validated instruments. Some of it is listed in Part 8. Secondary traumatic stress in child sexual abuse material (CSAM)-adjacent work is well documented in digital forensics and law enforcement populations. And the 2022 RAND study on trauma in the U.S. Intelligence Community is the closest structural analog to the broader mapping this table attempts.

What that literature almost entirely doesn't do is distinguish between the four injuries. It measures burnout, in populations that are also carrying secondary traumatic stress, vicarious trauma, and moral injury, using instruments built to detect one of them. That's the gap this framework is arguing about, and it means the cybersecurity evidence base can confirm that these roles are damaging people without telling you which injury is doing the damage in which seat.

So the mapping here is derived from established injury mechanisms applied to known role exposure profiles. That's a defensible methodology, and it should be read as clinical inference rather than empirical finding. A 2025 peer-reviewed workshop paper confirmed the shape of the gap directly, noting that cybersecurity roles were not explicitly identified in early information technology (IT) burnout studies and that empirical study of burnout by role remains sparse (Rangarajan et al., 2025). This framework is a starting point, not a settled body of evidence.

A note on vicarious trauma specifically: The VT entries in the table are a prediction, not a documented finding. Burnout has survey evidence. Secondary Traumatic Stress has well-documented analog populations in digital forensics and law enforcement. Vicarious trauma in cybersecurity has neither, but what it does have are all the precursors for it as seen in other fields. The mechanisms, the exposure profiles, and the practitioner reports that sound like VT are there when you know what to listen for. The framework names it anyway because someone needs to, and because the research needs to happen.

What This Map Is For

This map shows what's most likely, not what's possible. The table tells you where each role's exposure concentrates, not where it stops. Any role can produce any injury. A SOC analyst sits at a burnout floor, but the same seat can produce secondary traumatic stress or moral injury too. None of these are mutually exclusive. Read the table as a starting probability, not a ceiling. The injury you don't expect is the one that goes unnamed the longest.

The role-to-injury map is not a diagnostic instrument. It does not tell you which individuals on your team are injured. It tells you which seats are structurally exposed, which injury types to watch for in those seats, and where to concentrate your cadence investment before someone is already in the orange. The Chief Information Security Officer (CISO) who looks at this map and asks "which of these roles do I have, and what am I currently doing differently for them" is asking the right question.

The Role-to-Injury Map

The roles below are grouped by what drives their exposure. These injuries are not mutually exclusive, and the groupings describe the dominant pattern, not the only one. Follow the exposure and the symptoms, not the org chart.

Role	Primary Exposure Driver	Highest Injury Risk	Secondary Risk
CONTENT-FACING — direct exposure to traumatic material			
Dark web / CSAM-adjacent analyst	Content type	Secondary traumatic stress	Vicarious trauma
Cyber law enforcement / Internet Crimes Against Children (ICAC)	Content type + sustained load	Secondary traumatic stress	Vicarious trauma
CTI / threat intelligence analyst (including attribution work)	Content type + decision weight	Secondary traumatic stress	Moral injury
Malware analyst / reverse engineer	Content type + sustained load	Burnout	Secondary traumatic stress
DECISION-AND-OVERRIDE — judgment subject to non-technical override			
Ransomware negotiator	Decision weight + institutional friction	Moral injury	Burnout
Vulnerability researcher	Decision weight + institutional friction	Moral injury	Burnout
Operational Technology (OT) / Industrial Control Systems (ICS) practitioner	Decision weight + content type	Burnout	Moral injury
SUSTAINED OPERATIONAL LOAD — tempo-driven, burnout-primary			
SOC tier 1 analyst	Sustained load + decision fatigue	Burnout	Secondary traumatic stress

DFIR responder	Sustained load	Burnout	Vicarious trauma
Threat hunter	Sustained load	Burnout	Vicarious trauma
Incident commander	Sustained load + decision weight	Burnout	Moral injury

Many roles are not listed here. That does not mean they are safe. This table focuses on the roles whose exposure is distinctive enough to map specifically.

The Bullseye

Dark web collection and CSAM-adjacent work carries the highest single-role STS risk in the industry. The nearest documented analog population is digital forensics analysts in law enforcement, and two studies frame the risk. In a sample of 433 personnel on internet crimes against children task forces, nearly one in four showed low compassion satisfaction alongside high secondary traumatic stress and burnout (Brady, 2017). A smaller qualitative study of seven UK analysts who grade this material daily found that all of them said the work takes a toll, with most describing hypervigilance, mistrust, disturbed sleep, and a cynical shift in how they saw the world. The analysts who'd been in the role longest described the most symptoms (Strickland, Kloess & Larkin, 2023), which is the vicarious trauma pattern from Part 1 turning up in a documented population. Practitioners across these populations describe material involving harm to children as different in kind from other exposure, and the framework treats it that way.

CTI analysts carry a comparable profile. Cumulative dose is high, and the role carries enough decision weight to put moral injury risk alongside STS.

Ransomware negotiation is the clearest cybersecurity example of a role whose conditions favor moral injury, and it's worth being precise about what that claim is. This isn't a ranking, and nobody has measured it. It's an observation that the role concentrates the two conditions the moral injury literature identifies: decisions that carry consequences for people outside the room, made under constraints the decision-maker didn't set and can't change.

That combination isn't unique to security, and the professions where moral injury is best documented all share it. Military personnel making rules-of-engagement decisions, which is where the construct originates (Litz et al., 2009). Clinicians allocating scarce resources. Child protective caseworkers making removal decisions. Veterinarians. Journalists and humanitarian workers who document harm they aren't positioned to stop. In each case the injury doesn't come from the difficulty of the work. It comes from being the person whose name is on a decision that someone else's constraints shaped.

Several security roles share that structure and none of them have been studied. Vulnerability research where severity gets renegotiated. Threat intelligence where a finding is softened before it ships. Trust and safety where policy overrides the analyst's read. Incident response where the report the customer receives isn't the one the team wrote. If you want to know whether a role carries moral injury risk, the question isn't how hard the work is. It's whether the person doing it owns the outcome without owning the decision.

SOC tier 1 concentrates burnout risk more heavily than any other seat on the map. The volume is high, the autonomy is low, the triage is repetitive and high-stakes, and analysts rarely see how any of it ends. That's four of the six areas where a job and a person come apart, stacked in one seat, and it's the one part of this map with a cybersecurity-specific research base behind it. The decision fatigue load degrades in ways that look like performance issues before they look like a stress injury.

PART 3 — THE CADENCE

The zones tell you what you're looking at. The exposure map tells you where to look first. The cadence is how you make sure you're actually looking, on a schedule, before something goes wrong, with enough historical data to recognize a trend when you see one.

Most teams don't have a cadence. They have an open door, and an open door is not a cadence. It's a passive system that depends on the person in distress to self-initiate contact, accurately self-assess their own zone, overcome the stigma and professional culture that rewards self-reliance, and trust that disclosure won't carry career consequences. Every one of those dependencies fails at exactly the moment they're most needed.

The US military recognized this decades ago and built the battle buddy system into combat and operational stress doctrine specifically because the chain of command is not always the right first contact for someone who is struggling. The battle buddy works because they share the same environment, the same exposure, and the same professional culture — which means they can recognize a change in someone without a formal process to prompt them. This framework borrows that logic. It is not a military system and it is not the battle buddy program. It is a peer-supported triage and support model for cybersecurity practitioners, built on the same underlying principle: the person most likely to notice that something is wrong is the person standing next to you doing the same work.

What follows is written in two tracks. Read the one that fits your role. If you are both — a manager who also has a peer responder relationship with someone on a different team — read both.

Peer Response: How to Help Someone Alongside You

The peer responder is not a clinician, a counselor, or a substitute for either. What they are is trained, present, and connected to a pathway — and that combination reaches people before anyone else does.

Most peers who notice that something is wrong with a colleague don't fail because they don't care. They fail because they never had the vocabulary to name what they were seeing, the training to know what to do about it, or a clear path to connect someone to help. The result is a lot of informal checking in that stays at the surface and misses the injury underneath entirely. This section gives you what that informal check-in was missing.

What You Have That the Manager Doesn't

You are alongside. You share the same work, the same exposure, the same culture, and frequently the same dark humor that signals when something has crossed a line. You will notice a change in someone before their output metrics move, before they miss a deadline, before anything surfaces in a one-on-one. That early signal is the most valuable thing in this framework and you are the only one positioned to catch it.

You also have access to conversations a manager can't have. A peer can ask how someone is really doing without it feeling like a performance review. A peer can sit with someone after a hard session without it meaning anything official. A peer can say I noticed you haven't been yourself since that engagement without it carrying career weight. That access is the foundation of everything that follows.

You don't have to be friends

Nothing in this role requires liking the person, or being liked by them. The peer responder track runs on proximity and shared work, not closeness. Someone you barely socialize with can still notice you've gone quiet, and sometimes that lands harder than concern from a friend. If a person you're not close to noticed, the change was visible enough that a non-friend saw it, and that is harder to wave off.

Your Entry Points

You don't have a calendar and you don't have a reporting line. What you have is proximity and relationship, and those create entry points that a manager's formal structure doesn't. The peer check-in doesn't look like a meeting. It looks like a moment.

- **Before standup or at shift change.** Two minutes before the team gathers. Low stakes, no audience, easy to normalize. How did the week land? Are you sleeping well? Are you still spun out from that thing last week?
- **After a hard call or a significant exposure.** The immediate post-event window is the highest-value moment the peer has. Not a formal debrief — just presence. That was a rough one, how are you dealing with it?
- **In the margins of normal contact.** Slack, a walk to the car, grabbing coffee. The peer check-in travels in whatever channel the relationship already uses. The key is that it happens with intention, not just when something is visibly wrong.
- **When you notice something has changed.** This is the most important entry point and the hardest to name. You don't need a specific event to justify checking in. A change in someone's tone, their humor, their engagement, their energy — that's signal enough. I've noticed you seemed a little different lately. Are you good? I'm here to help if you need it, I got you.

How to Build a Baseline Without a One-on-One

The manager builds a baseline through scheduled contact. The peer builds it through relationship over time. The mechanism is different but the goal is the same: know what this person looks like when they are okay, so you can recognize when they aren't.

Ask the questions when the answers are green. Not just when something seems off. Not just after a hard engagement. Regularly, in the margin of normal contact, so that the answers accumulate into a picture of what this person's normal actually looks like. That picture is your baseline. You are recording anomalies against it: when the answers start deviating from that normal, you'll notice, because you have something to compare against. Start recording from wherever you are now, even if that's not a good place.

What You Can't Do — and What To Do Instead

You cannot move someone's workload. You cannot change their rotation or their exposure. You cannot formally escalate within the organization without involving the manager. Knowing where your lane ends is as important as knowing what to do within it. Instead, consider the following:

- **Cover informally where you're able.** You can carry something for someone, take a task off their plate, or just sit with them after a hard session. Small acts of coverage are within your reach without any formal authority.
- **Flag your observations to the manager.** When what you're seeing is beyond peer-level support, the manager needs to know. You don't have to disclose what was said in confidence. You can say I've noticed something is off and I think it warrants a check-in from you.
- **Bridge to clinical support.** When the peer's lane ends, the handoff to clinical support is the most important thing you do. Make it warm. Stay present through it. Follow up after.

Manager Response: How to Help Those You Lead

The manager has structural tools the peer doesn't: scheduled access, formal authority, and visibility into output and attendance. The cadence guidance below is built around those tools. A calendar invite removes every dependency that makes open-door access fail. The contact is scheduled, it doesn't require self-initiation, it doesn't signal that something is wrong, and it's just the next check-in, the same way the last one was.

Those tools are worth more than they look. In a study of 600 personnel on internet crimes against children task forces, the strongest protective factor against secondary traumatic stress wasn't any coping strategy the individual used on their own. It was having a supervisor who supported them (Bourke & Craun, 2014). To the person doing the work, you are what the organization looks like.

One precondition comes before any of it. A manager can't read a team clearly through their own depletion. This isn't a failing; it's the same principle the rest of this document runs on. Nobody sees their own drift well, and that includes you. So your own baseline isn't separate from the job of watching your team's. It's part of it. Managers need what they're providing: someone above them, beside them, or outside the reporting line who's checking on them the way they're checking on their people.

Use What Already Exists

Before building anything new, prefer the meeting vehicles you already have. If there is already a recurring one-on-one, a shift-change handoff, or a standing team sync, the health check-in rides inside it rather than adding another meeting. If nothing suitable exists, build it holistically, integrating the operational conversation and the health check-in into one contact rather than bolting wellbeing on as a separate, easily-deprioritized event. The goal is a rhythm the team will actually keep, not a program that looks good on paper and gets cancelled the first busy week.

Know What the EAP Provides Before You Refer

Before you route anyone to the Employee Assistance Program, know what that specific EAP actually provides. EAPs vary widely, and a manager who refers someone to a generalist EAP that isn't equipped for the injury in front of them has not solved the problem, only relocated it. Find out the session limits, whether the counselors have any trauma-informed or occupational-trauma training, and how referral versus self-initiated contact affects confidentiality (Part 5 covers why that distinction matters). Refer with knowledge, not on faith.

Establish the Baseline at Green

No detection capability works without a baseline. A security information and event management (SIEM) platform without historical data isn't detecting anomalies, it's generating noise. The delta is the signal, and you can't identify a delta without a reference state to measure against. The same principle applies to practitioner monitoring. A single check-in tells you where someone is today but tells you nothing about direction of travel, rate of change, or how far they've moved from where they started. The peer responder who only has current zone state is running a detection system with no historical data. They can see what's in front of them but they can't see the trend line. This is why the baseline has to be established at green, before the surge, before the incident, before anything has shifted.

Cole's drift was invisible to everyone around him not because nobody was paying attention, but because nobody had a record of who Cole was at year two. By year eleven there was nothing to compare against, and the detection failed because the baseline was never captured. Ask the questions when the answers are green so you know what green looks like for this specific person, because green looks different for everyone, and generic wellness questions answered during a surge are measuring the wrong thing against the wrong reference point.

What "Recording" Means Here

Everything above borrows its language from detection engineering, and it's worth being literal about where the metaphor stops. The baseline you're building is something you hold, not something you file. Don't keep notes on colleagues.

Part of that is about trust. A peer responder who takes notes is doing something different from a peer responder who pays attention, and the person on the other side of the conversation will feel the difference. The moment someone suspects they're being documented, you've lost the thing that made the check-in work.

The rest is practical. Informal notes about a named colleague's wellbeing are sensitive personal data in most jurisdictions, they carry no privilege, and they're discoverable in any subsequent employment dispute. In the EU and UK they may fall under the special category rules in data protection law. In the US, an employer accumulating a wellbeing record on an employee raises its own questions. None of that is a reason to pay less attention. It's a reason to keep what you notice between you and the person.

If an organization decides it wants a formal record of any of this, whether that's an assessment cycle, an escalation log, or a program of any kind, that's a legitimate choice and a different one. It belongs with counsel and a privacy review before it starts, not after.

Re-Baseline After Significant Events

In network monitoring, you periodically re-baseline when the environment has legitimately changed: new infrastructure, new traffic patterns, planned architectural shifts. The old baseline is no longer valid, and comparing against it will generate false positives or miss real anomalies. The same applies to practitioners. Someone who has just rotated off a high-exposure role, closed a major engagement, experienced a significant personal stressor, or returned from extended leave has a legitimately shifted baseline. Significant events that warrant a re-baseline check-in include: engagement or incident close after sustained high-tempo work, role change or significant shift in exposure profile, return from extended leave, team loss through attrition or reorganization, and known personal stressors outside work. The re-baseline check-in is not a wellness flag. It's a calibration event. A study of former NSA tactical cyber operators found that sleep disturbance persisted for years after operators left the role, with none citing pay as the reason for departure, suggesting that the psychological load of the work outlasts the work itself (Rangarajan et al., 2025).

What Other Fields Have Found Workable

The Professional Quality of Life Scale (ProQOL), Stress First Aid, and Combat and Operational Stress Control (COSC) stress continuum doctrine share one operational assumption: structured, leader-initiated contact at defined intervals outperforms passive open-door access on every wellbeing outcome measured. Voluntary help-seeking in high-stigma, high-self-reliance professional cultures is a poor detection mechanism. Scheduled contact is not. What follows is drawn from the cadences those fields use. It's offered as a starting point to adapt, not as a specification to meet, and none of it needs a budget, a program, HR involvement, or clinical training to try.

- **Weekly during active surge or high-exposure work.** Not a formal check-in but a two-minute conversation before standup. How are you holding up. Are you sleeping. Anything from this week still running in the background. The specific questions matter less than the regularity, because regularity is the signal that someone is paying attention.
- **Bi-weekly during normal operational tempo.** Fifteen minutes, scheduled, recurring, and explicitly not performance-focused. Keeping them separate is one of the most important structural decisions a team lead can make, because conflating wellbeing with performance is one of the fastest ways to kill disclosure.
- **Soon after a significant exposure event.** Don't wait for the person to raise it. A specific identified high-dose exposure is worth following up on regardless of how the person presented in the moment. Someone can walk out of it looking fine and not be fine by the weekend.
- **At engagement close.** This is the yellow-orange diagnostic moment. The surge is over, and the question is whether they returned to baseline. Schedule it before the engagement ends so it doesn't get deprioritized by whatever comes next.

What the Cadence Is Actually Doing

A structured cadence does three things an open door cannot. It captures longitudinal data, so that a manager who has been checking in every two weeks for six months has a comparison baseline and knows what this person's normal looks like. It normalizes the conversation, so that a check-in happening on a schedule removes the signal value of the check-in itself and creates the environment in which disclosure becomes possible. And it removes the activation energy, because the person in yellow or orange is not going to self-initiate, they are going to rationalize, minimize, and wait for it to pass. The intelligence community has the same problem. The same RAND study that named secrecy as a risk factor also found that supervisors there aren't necessarily trained or equipped to recognize secondary traumatic stress, and that the people experiencing it may not have the vocabulary to describe it well enough to ask for help (Sudkamp et al., 2022). Neither of those is fixed by an open door. The cadence is the intervention.

A Note on Frequency and Role

The cadence above applies to all high-exposure roles. For roles at the top of the exposure map, specifically dark web and CSAM-adjacent analysts and ransomware negotiators, it probably isn't enough on its own. What tends to get added for those roles: more frequent contact, often weekly. Peer responders with specific training in secondary traumatic stress and moral injury. Rotation schedules that treat high-exposure work as a limited-duration assignment rather than a permanent function. And some form of longitudinal measurement, since cumulative drift doesn't surface in conversation. The cadence doesn't replace clinical support for those roles. It's the early warning system that gets people to clinical support before the injury is severe enough to require it.

PART 4 — HOW TO CHECK IN

Knowing the cadence tells you when to show up. This section tells you what to do when you get there.

The peer responder role sits in a specific and bounded lane. It is not therapy, not clinical assessment, and not crisis intervention. It is structured peer support delivered by someone who knows the work, knows the person, and has agreed to pay attention. The value of that role is precisely its informality. The risk of that role is scope creep in both directions: the peer who tries to treat what they're seeing, and the peer who pulls back because they're afraid of saying the wrong thing. Both failures leave someone in the orange without support. The check-in is not a difficult conversation. It's a short, recurring, low-stakes contact that accumulates into detection capability over time. The goal is not to fix anything. The goal is to see clearly, name what you see, and know when to hand off.

The Seven C's: What You're Actually Doing

Stress First Aid gives the peer responder seven functions, not seven steps. They are not sequential and not a script. They are a set of actions a non-clinical peer can deliver in any order the conversation requires, without a license or a clinical background.

Check Assess where they are and how they've moved since the last contact. This is your delta measurement. Questions should be consistent enough across check-ins that you can actually compare the answers. That consistency is what turns a conversation into longitudinal data.

Coordinate Determine the trajectory. Is this getting better, staying the same, or getting worse since the last check-in? That single assessment is your severity call. It tells you whether to stay in the respond lane or move toward escalation.

Cover Reduce the load where you can. Move a task. Adjust exposure. Push a deadline. This is where the peer responder's organizational position matters. You can't cover for someone you have no relationship with and no standing to help.

Calm Reduce acute distress without minimizing what the person is carrying. The calibrated response is to name what you're observing without diagnosing it and without attaching more weight to it than the person has given it themselves.

Connect Maintain the relationship and counter the isolation that compounds every injury. Connection is not a deep conversation every time. It's the consistent signal that someone is present and paying attention.

Competence Restore the person's sense of their own capability and the value of the work. This is not a pep talk. The distinction is specificity. "You're a good analyst" is a pep talk. "The way you handled that attribution call was exactly right" is a competence intervention.

Confidence Build the person's confidence that things can improve and that support is available. This means being concrete and specific about what the next step is and following through on it. Credibility is built by follow-through, not by words.

A note on using the C's on yourself

The C's are built for checking on each other, and that's where they do the most good. But there's nothing stopping you from turning them inward. If running through them for yourself helps you notice where you are and process what you're carrying, use them that way too. Just know that the further you slip, the harder your own read gets, which is exactly why the peer version exists.

When the work can't be discussed

Part 2 names secrecy as an exposure driver. Even with information sharing limitations, here's what still works once the wall is up. The zones disclose nothing, because "I'm in the orange this week" is a statement about state, and state isn't secret. Check, Coordinate, Calm, Connect, and Confidence all work without knowing the cause. Two things don't work, Cover and Competence. You can't cover work you aren't read in on, though you can take the uncleared load so there's room for the rest, and Competence needs specifics you may not have. A peer outside the wall can affirm the person. They can't affirm the work.

The failure is treating "I can't talk about it" as the end of the conversation. Even if the content is off limits, the zone state isn't. Ask the second question anyway.

Also, understand that this framework will struggle to support a party of one. Someone who is isolated and unable to connect with anyone at all due to secrecy is running a higher risk of stress injury, and the limited protective mechanisms become even more constrained.

The Seven C's in Practice

Priya — CTI Analyst — the peer responder

This is the same incident from Part 1, seen from the other side. Marcus is the analyst who took the pediatric-oncology disclosure as a secondary traumatic stress injury. Priya is the colleague who noticed.

Priya noticed something was off with her colleague Marcus about a week after the pediatric oncology disclosure. Not often enough that anyone would have flagged it. He was still showing up and still closing tickets. Then he just didn't seem like his normal self, a bit more withdrawn than usual. He stopped making his usual jokes at standups and wasn't really talking much on Slack.

She caught him before standup on a Tuesday.

Check. "Hey, you doing okay? You've seemed a little different lately." Marcus shrugged. Said he was fine, just tired. She didn't push. She noted the answer and filed it against what she knew from three months of working alongside him. It wasn't his normal tired.

Coordinate. She checked in again Thursday after a routine threat briefing. Same flatness, maybe slightly worse. The trajectory wasn't improving. That told her something.

Cover. She picked up two of his open monitoring tasks that week without making a thing of it. Told him she had bandwidth and he could grab them back whenever. He didn't push back, which told her more.

Calm. She didn't tell him what she thought was happening. She didn't name the disclosure or ask him directly about it. She sat with him after Thursday's briefing and said it had been a heavy few weeks. He nodded. That was enough.

Connect. She kept showing up. Not with an agenda. Just present, consistent, in the margins of the normal workday. The same way she always had been. That continuity was the signal that nothing had changed between them.

Competence. On Friday she mentioned in passing that the collection report he'd filed on the disclosure was thorough and clean under very difficult conditions. It wasn't a pep talk, but a very specific and true thing about work he had done very well.

Confidence. When she finally said "I think you should talk to someone, and I'll sit with you while you make the call," he didn't push back on that either. She had built enough of a track record in the previous two weeks that the offer landed as something real rather than something said to make herself feel better.

She wasn't his manager, but she was a peer who understood her coworker well and understood Stress First Aid. She made a point of checking in with him and being a presence for him, and she used what she had: two weeks of paying attention and the training to know what to do with it.

What a Check-in Actually Sounds Like

In practice a fifteen-minute check-in with someone in yellow or early orange sounds less like a structured interview and more like a conversation between two people who know each other and work in the same field, where one of them is paying slightly more careful attention than usual. What matters is listening for three things: content, which is what they're actually describing; tone, which is whether how they sound matches what they're saying; and delta, which is whether this is different from last time. You are not diagnosing. You are comparing current zone state to prior, noting the delta, and deciding what the delta means for your next move.

The Boundary

The peer responder's lane ends in three places: when someone is in the red zone; when someone is in orange and not moving after a couple of structured check-ins; and when the content of the disclosure exceeds the peer's capacity to hold. The peer who absorbs disclosures beyond their lane without escalating is not helping. They are taking on a load that will move them toward their own orange zone while the person they're trying to support remains without appropriate care.

What Not to Say

Every one of these gets said by people who mean well. They're what fills the gap when someone wants to help and doesn't have the right language for it, which is the gap this whole document exists to close. The problem isn't the intent, it's that each one ends the conversation with no meaningful relief. This isn't an exhaustive list, just examples to help you hear the difference and respond in a healthier, more productive way.

"You need to toughen up." *resilience myth.*

Telling a burned out analyst to be more resilient does nothing to address the cause of the burnout or the dosage.

"Everyone feels this way sometimes." *normalization.*

Normalizing is not the same as acknowledging. It signals that the disclosure has been classified as routine and dismissed.

"Have you tried taking a long weekend?" *wrong intervention.*

Rest prescribed for the wrong injury closes the alert without resolving the incident.

"I'm sure it'll pass." *false reassurance.*

If they are in orange, it will not pass on its own. That's the operational definition of orange.

"Why don't you clock out early and go have a beer?" *substance as coping.*

A very common thing to hear, in a lot of workplace cultures. It points someone toward a substance as the way to manage what they're carrying, and offers relief that doesn't touch the injury underneath.

"We have mental health support on our insurance." *false enablement.*

Naming that a benefit exists is not the same as connecting someone to it. It puts the entire burden of initiation on the person least equipped to carry it, and treats a phone number as if it were support.

"That's a you problem." *character attribution.*

It recasts the stress injury as a personality issue. Nothing about the situation changes, the exposure stops being anyone's responsibility, and they're left holding it alone.

"You should talk to Human Resources." *deflection.*

Hands the person off instead of supporting them, toward a channel that isn't built for this. (See callout.)

What Human Resources is for, and what it isn't

HR isn't a confidential clinical resource, and it isn't a substitute for one. What you disclose there is generally a business record. That's worth knowing before you disclose, and it's a reason to reach clinical support directly rather than through HR.

But HR isn't a channel to avoid either, and saying so would do readers a disservice. In many jurisdictions it's the correct and sometimes the only route to workplace accommodations, medical leave, schedule or workload changes, and role adjustment. In the United States those requests are what open an employer's obligations under the Americans with Disabilities Act. Elsewhere the route may run through occupational health, a works council, or a statutory equivalent.

Two different functions. Clinical support is for the injury. HR is for the job. Someone may need both, and the sequencing is theirs to choose. What this framework asks is that nobody is routed to HR instead of support, and that nobody is steered away from an accommodation they're entitled to.

PART 5 — ESCALATION

The peer responder is a first responder, not a clinician. The cadence and the check-in protocol exist to detect early and intervene before injuries become severe, but they also exist to create a clear and practiced path to higher-tier support when the peer's lane ends. Escalation is not a failure of the peer responder role. It is the role working exactly as designed. The failure mode is not escalating too quickly. The failure mode is escalating too late, or not escalating at all.

IF SOMEONE IS IN IMMEDIATE DANGER — CALL 911*

IF SOMEONE IS IN CRISIS — CALL OR TEXT 988*

These are not last resorts. They are first responses. Use them.

* US numbers. Outside the US, see the international crisis lines in the resources section.

When to Escalate

- **Red zone presentation.** Weeks of worsening symptoms, real functional impairment, hopelessness, or any indication that the person is a risk to themselves. The peer responder who is looking at red does not have a support conversation. They have one job: bridge to a professional and stay present until the bridge is made.
- **Orange with no trajectory improvement.** The protocol has been run, the load has been reduced where possible, and the person isn't recovering. Two check-ins is a reasonable point to stop and reassess, and it's roughly where the peer-support literature suggests a non-clinical responder has done what they can. Two check-ins is enough time to distinguish a temporary plateau from a genuine stall, and short enough that a person in orange isn't sitting without escalated support for weeks.
- **Disclosure that exceeds the peer's capacity.** This is the boundary that peer responders most often fail to recognize in themselves. The peer who absorbs disclosures beyond their lane without escalating is not helping. They are taking on a load that will move them toward their own orange zone.

The Warm Handoff

A referral is not a handoff. Sending someone a link to an EAP or a phone number is a referral. It places the entire burden of initiation on the person least equipped to carry it. Warm handoffs have meaningfully better completion rates because they don't depend on self-initiation at the moment self-initiation is hardest. A warm handoff means the peer responder stays present through the transition, describes what they've observed in plain non-clinical language, and follows up after to confirm the handoff landed and that the person is actually being supported.

EAP: What It Is and What It Isn't

Employee Assistance Programs typically provide short-term counseling, commonly three to eight sessions though it varies by contract, with a referral to longer-term care if needed. They are not specialized for occupational trauma, secondary traumatic stress, or moral injury unless the organization has specifically contracted for those capabilities, which most have not. A generalist EAP counselor receiving a practitioner who has been doing dark web collection for three years is encountering a clinical presentation they were almost certainly not trained for. That mismatch is a real limitation on what the EAP can do for the injuries this document describes.

Two things follow from this. First, the EAP is a starting point, not a destination, and it should be supplemented with more specialized resources when the presenting injury requires it. Second, there's usually a confidentiality distinction that changes which path is safer. In the typical arrangement, self-initiated EAP contact is confidential and the employer receives only aggregate usage data rather than names or content. A manager-mandated or formally-referred contact often isn't confidential in the same way, and the employer may receive confirmation of attendance or completion. Arrangements vary by provider, contract, and jurisdiction, and every EAP has limits on confidentiality where there's risk of harm. Nobody should assume how theirs works. The principle holds regardless: the safer path is usually the one the person reaches themselves, rather than through a formal referral that creates a record.

The Follow-Up Obligation

Escalation is not the end of the peer responder's involvement. The follow-up cadence after escalation should be lighter than the pre-escalation cadence, because the clinical support is now doing the primary work. But it should be present, scheduled, and explicitly framed as separate from the clinical support. The peer responder who stays on the thread after escalation is the safety net for premature discontinuation, wrong-fit resources, and disengagement from support without anyone noticing.

One caution on what following up can and can't look like. When you've handed someone off to a clinical resource, you might want to know it took. You can't get that from the resource. A therapist or EAP counselor can't confirm to you whether someone showed up, or what was said, and that's not a wall to work around. It's the same confidentiality that made the handoff worth making. So the follow-up goes to the person, not the provider. Not "did you go," but "how are you doing, are you getting what you need?"

Escalation Is Not Surveillance

The peer responder framework depends on trust, and trust depends on the person being supported having confidence that check-ins and escalations are not being reported to management, logged as performance concerns, or used to make decisions about their role or career. The peer responder who escalates to a manager rather than to a clinical resource, or who logs check-in content in a performance system, has not run the protocol. They have broken the system that makes the protocol possible. The only exception is imminent risk to the person or others, which overrides confidentiality in the same way it does in every professional support

context. That exception should be named explicitly at the outset of the peer responder relationship.

PART 6 — THE ORG LAYER

Everything in this framework up to this point operates below the level of the organization. All of it is necessary, and none of it is sufficient if the organization itself is generating conditions that produce injury faster than peers can detect and respond to it.

Who holds up the people holding others up

An organization can't put the whole early-warning system on its managers and assume the managers stay healthy. Managers carry the same exposure as the people they lead, sometimes more, and a manager who goes down quietly takes the team's visibility down with them. The same is true of peer responders: the person absorbing everyone else's hard sessions is carrying a load of their own. Building support that assumes the manager or the peer is the permanent, reliable sensor is building a single point of failure. The people who hold others up need someone holding them up too, and that's the org's job to provide, not the manager's or the peer's job to arrange alone.

What Organizations Must Stop Doing

- **Treating high-exposure roles as entry-level positions.** The SOC tier-1 analyst doing triage on graphic content, the junior CTI researcher assigned to dark web collection, the new DFIR associate handling victim notification calls. These are high-exposure roles that carry high injury risk. High caseloads in child protective services have been associated with high levels of STS, and the parallel in cybersecurity is a tier-1 SOC analyst carrying an alert volume that structurally prevents the kind of deliberate, supported processing that high-exposure work requires.
- **Relying on individual resilience as the primary intervention.** The six positive fits that protect against burnout: sustainable workload, choice and control, recognition and reward, a supportive work community, fairness and respect, and clear values and meaningful work. These are organizational, not personal factors. Resilience training, wellness apps, and mindfulness programs address none of those six domains. An organization that invests in individual coping resources while leaving workload, control, reward, community, fairness, and values mismatches intact is treating the symptom and preserving the cause.
- **Eliminating or under-resourcing supervision for high-exposure roles.** In child welfare and clinical social work, structured supervisory contact that creates explicit space for practitioners to process the emotional and moral weight of the work is a documented protective factor against STS and vicarious trauma. A team lead whose one-on-ones are exclusively output-focused is providing performance management, not supervision. Those are different functions and they do not address the same need.
- **Creating climates where disclosure carries career risk.** An organization where practitioners believe that disclosing psychological distress will result in reduced responsibility, reduced trust, or reduced career opportunity has made the peer responder framework

structurally impossible to run. Climate change of this kind requires active visible commitment from senior leadership, not policy statements.

- **Compartmentalizing the work without providing support inside the compartment.**

An organization that decides who is allowed to know what has also decided who is available to help. That's a consequence it owns, not a side effect. The people it isolates need a channel that reaches them where they are: cleared or privileged clinical support, a peer arrangement covered by the same agreement, someone senior who is read in and whose job is the person rather than the project. Routing someone to an Employee Assistance Program they can't legally be honest with isn't support. It's a referral to a wall.

- **Treating attrition as the cost of doing business.** Attrition in high-exposure cybersecurity functions is often treated as an inevitable consequence of the work's difficulty rather than as a measurable outcome of organizational conditions that could be changed. The practitioner who leaves after two years in a dark web collection role is not simply a retention statistic. Their departure is a signal that the exposure management, supervisory support, rotation policy, and organizational acknowledgment of the role's risk may have been insufficient. Aviation's Fatigue Risk Management System, which treats crew rest as a structural safety requirement rather than a personal wellness choice, has been proposed as a direct analog for high-exposure cybersecurity roles (Rangarajan et al., 2025). And for roles carrying vicarious trauma risk specifically, the injury does not stay contained to the practitioner. The worldview drift moves outward into the people closest to them: partners, families, friendships. That pattern has been described in other professions that carry indirect trauma, and there is no reason to think this one is exempt. The injury you are not treating does not stay in the building.

A Starting Point for Organizational Self-Assessment

Part 1 made the case that these four injuries are not interchangeable, and that calling all of them burnout is how people spend six months on the wrong intervention. The same trap is waiting for organizations. There's no single score that tells you whether your organization is hurting its people, and if you only assess the injury you happen to have a tool for, you'll miss the other three.

Five questions cut across all four injuries. Start with these, because you can answer them today, without a budget line, a vendor, or anyone's permission.

- Does your organization acknowledge secondary traumatic stress and vicarious trauma as occupational hazards in writing, in policy, and in the hiring process for high-exposure roles?
- Do supervisors of high-exposure teams have explicit training in recognizing and responding to STS and moral injury?
- Are rotation schedules and exposure limits for high-content roles formally defined or left to individual discretion?
- Are there protected pathways for practitioners to surface psychological distress without career consequences?

- Does your organization measure practitioner wellbeing longitudinally, or only when someone flags a concern?

If the honest answer to most of those questions is no, the organization has identified its starting point. The peer responder framework, the cadence, and the check-in protocol are all more effective in an organization that is also working on those structural conditions. But they are not useless in an organization that hasn't addressed them yet. They are the intervention that can be running while the structural work is underway, the detection and support capability that catches people while the organization is still learning how to stop producing injury at the rate it currently does.

Assessing Each Injury

Past those five questions, each injury has its own assessment picture, and the four are in very different shape. What follows takes them in the order Part 1 introduced them. Three have an instrument that scores the organization instead of the person. One has nothing at all.

Burnout

Burnout is the best-served of the four, and the only one where the organizational question already has a mature answer. The Areas of Worklife Survey, from Leiter and Maslach, scores the same six areas listed earlier in this Part: workload, control, reward, community, fairness, and values. It works at the level of a single team or as a summary profile across a whole organization, and run alongside the Maslach Burnout Inventory it tells you both how much burnout is present and which of the six mismatches is producing it. That second half is the part that matters, because it's the difference between knowing your DFIR team is depleted and knowing whether it is the tempo, the lack of control over what lands in the queue, or the fact that nobody's ever acknowledged what the work costs.

Both instruments are commercial, licensed through Mind Garden. If cost is the obstacle, the Copenhagen Burnout Inventory is validated, peer-reviewed, and free to use, though it measures individuals rather than the organization around them.

Knowing which area is failing is only half of it. The Job Demands-Resources model sharpens what to do next: demands drive exhaustion, and resources, meaning autonomy, support, recovery, and recognition, buffer the damage (Demerouti et al., 2001). Because the demands in high-exposure security work are often intrinsic and cannot be removed, the organizational lever is resources: rotation, peer support, and supervision are not 'perks' but the buffer that keeps an unavoidable demand from becoming a stress injury.

Secondary Traumatic Stress

The Secondary Traumatic Stress Informed Organization Assessment (STSI-OA) is a 40-item instrument that scores a workplace, not a person, on whether its policies, supervision, workload practices, and culture protect against secondary traumatic stress or accelerate it. It was developed by Ginny Sprang and colleagues at the University of Kentucky Center on Trauma and Children, is distributed through the National Child Traumatic Stress Network, and is free to use.

It was built for child welfare and would need real adaptation, so this framework does not propose that security organizations administer it as written. What it does propose is that the domains it assesses are the right domains to interrogate, which is what the five questions above are drawn from.

Not every security organization carries meaningful secondary traumatic stress risk, and this framework does not claim otherwise. A team building detections and tuning a SIEM is not carrying what a forensic examiner is carrying. The role-to-injury map in Part 2 is where to check. If your organization staffs dark web collection, CSAM-adjacent forensics, ransomware negotiation, DFIR, or CTI, then you have seats where secondary traumatic stress is a structural feature of the work rather than a possibility. Those are the organizations the questions above are written for, and that is why you are reading this paragraph.

The relevance isn't that cybersecurity resembles child welfare. It's that both are populations exposed to other people's trauma, indirectly and repeatedly, as a condition of employment. That is the exposure profile the instrument was built to assess, and it is the one those roles carry. Same borrowing logic as taking the stress continuum from Marine Corps doctrine without claiming security work is combat. No one has administered the STSI-OA to a cybersecurity organization, so there is no validation in this population. What exists is a validated instrument asking the right structural questions of a comparable exposure profile, which is more than this field has built for itself, and its value is that it converts a vague sense that the organization might be part of the problem into something specific enough to act on.

Vicarious Trauma

The Vicarious Trauma Organizational Readiness Guide (VT-ORG) assesses an organization's capacity to address its people's work-related exposure to trauma, across five areas of organizational health. It was produced by Northeastern University for the US Department of Justice Office for Victims of Crime, it is free, and it ships in separate versions for victim services, emergency medical services, fire services, and law enforcement. A companion Blueprint walks a leadership team from the results through to a plan, and then back to the instrument later to measure whether anything actually moved. It's also been validated: across 3,018 employees at 13 first responder and victim assistance agencies it showed excellent internal consistency, and its subscales predicted turnover intention, compassion satisfaction, and organizational resilience (Hallinan et al., 2019).

The re-administration step is why this one matters. Vicarious trauma is the injury that hides from single-point detection, and Cole's drift went unnoticed because nobody kept a record of who he was at year two. An instrument built to be run twice is the organizational version of the baseline this framework keeps asking practitioners to build.

Moral Injury

Nothing. There's no organization-level instrument for moral injury. Measures do exist, but they assess individuals, and most of them were built for military populations. Nothing scores whether an organization is producing moral injury in the people who work for it.

Until something does, the closest this framework can offer is its own Part 2. Decision weight and institutional friction are the two exposure drivers that generate moral injury, and both are conditions an organization can look at directly. How often does professional judgment get overridden for reasons that have nothing to do with the evidence? Who carries the cost when it does? Has anyone ever named that cost out loud? That's not an instrument, it's a starting point, and more than this field has now.

The Evidence Behind This

The arguments above stand on established instruments and doctrine. For readers who want to operationalize them, or who need the citations to make the case internally, here is what they rest on.

Source Bodies: Why These, Why Here

The instruments for each injury are described above. Two source bodies sit under the whole framework rather than under any one injury, and they are worth knowing for themselves.

COSC — Combat and Operational Stress Control (MCTP 3-30E, MCO 5351.1A)

The US Navy and Marine Corps doctrine for managing psychological health under sustained operational stress. Developed over two decades, COSC establishes the stress continuum model used throughout this framework, the five core leadership functions of strengthen, mitigate, identify, treat, and reintegrate, and the Combat and Operational Stress First Aid protocol from which the 7 C's are derived. Cited here not only as a vocabulary source but as an organizational model: COSC is explicitly commander-accountable, mandated at every level of the chain of command, and built on the premise that psychological health is an organizational responsibility. That premise is the one this framework argues cybersecurity organizations need to adopt.

ProQOL — Professional Quality of Life Scale

Developed by Beth Stamm and freely available through the ProQOL organization. One of the most widely used and validated measurement instruments for compassion satisfaction, burnout, and secondary traumatic stress across helping professions. Designed for repeated administration to capture longitudinal change rather than point-in-time state. Recommended in the cadence section for 90-day assessment cycles in high-exposure roles.

That's the foundation this framework is built on. The harder question, and the one a CISO will ask, is whether any of it actually changes anything.

Does Any of It Work?

It does, and somebody went and measured it. Child welfare staff who describe their workplaces as supportive report less secondary traumatic stress. More usefully, a longitudinal study followed organizations through deliberate change and found that the ones whose STSI-OA scores improved saw significant declines in secondary traumatic stress, and the result survived controls for age, gender, time, and how much implementation work each organization actually did. Burnout moved too, but only in the organizations doing the most implementation work, which is its own finding about half measures (Sprang, Lei & Bush, 2021). The organization

moved first, and the people got better. Where the organization doesn't move, the injury accumulates regardless of what individual practitioners do to manage it.

Burnout research has spent decades narrowing down where a job and a person come apart, and it lands on six places: workload, control, reward, community, fairness, and values (Leiter & Maslach, 2003). Most difficult jobs fail one or two of them. High-exposure security roles fail all six, and not by accident. This is the job.

Where it comes apart	What that looks like in security work
Workload	Incident tempo in CTI and DFIR that no rotation absorbs
Control	Analysts in collection or triage don't choose what they see, or for how long
Reward	The highest-exposure seats staffed as entry-level positions
Community	Work that can't be discussed outside the team, or outside the building
Fairness	A finding softened, a severity renegotiated, or a disclosure decided on business grounds
Values	Institutional friction overriding professional judgment

Read that table as a list of things an organization chose, because every row is a decision someone made and could make differently.

Further Evidence

The rest of the case comes from three different directions, and they all arrive at the same place.

Start with doctrine, which makes it somebody's job. The US military figured this out early: COSC requires commanders at all levels to fully execute and sustain an active program regardless of structure, size, or location. The CISO or people leader who says their door is open isn't meeting that standard. The standard is an active, structured, leader-executed program.

Then there is the question of which interventions actually pay off. A 2024 systematic review of organisational interventions in healthcare found that the approaches with the clearest benefit were the ones that changed working conditions, not the ones that trained individuals to cope with them (Aust et al., 2024). The same research group has since cautioned that those gains fade without sustained organisational commitment, which is its own argument: this is a standing responsibility, not a project. Resilience training isn't useless, it's just the smallest lever and it's one that organizations often reach for first.

The US Surgeon General says the same thing. The 2022 Framework for Workplace Mental Health and Well-Being was written for every workplace in the country, and it organizes the argument around five essentials: protection from harm, connection and community, work-life harmony, mattering at work, and opportunity for growth. Read that list against a SOC floor or a DFIR on-call rotation and count how many you'd actually score. None of this is fringe, and none of it is specific to cybersecurity. The rest of the working world is just further along than we are.

IN CLOSING

The goal of this framework is to make the cybersecurity practitioner space a kinder, more caring, and healthier place to work.

As you read this document, you may have found yourself thinking about your own hard moments, and about people you have watched struggle, or reach a crisis, without the words to name what was happening. That response is normal, and expected. It was part of the catalyst for building this framework in the first place.

Finding the right words for something is a real turning point. You now have more of those words, and more resources, to go further in your own wellbeing and in helping the people around you.

Be kind to yourself and to others, and keep fighting the good fight.

PART 7 — CRISIS AND SUPPORT RESOURCES

This section is here so it can be found fast. If you or someone alongside you needs help right now, start at the top. The crisis lines are immediate and confidential. Below them are the training and support resources this framework refers to throughout.

IF SOMEONE IS IN IMMEDIATE DANGER — CALL 911*

IF SOMEONE IS IN CRISIS — CALL OR TEXT 988*

These are not last resorts. They are first responses. Use them.

* US numbers. Outside the US, see the international crisis lines below.

Crisis Lines

988 Suicide and Crisis Lifeline. Free, confidential, 24/7. Call or text 988. English, Spanish, and 240+ languages. Dedicated support for Deaf and hard of hearing callers, including ASL videophone, at <https://988lifeline.org/deaf-hard-of-hearing-hearing-loss>. <https://988lifeline.org>

Veterans Crisis Line. Call 988 then press 1, or text 838255. Free, confidential, 24/7. For veterans, service members, and their families. <https://www.veteranscrisisline.net>

Crisis Text Line. Text HOME to 741741. Free, confidential, 24/7. <https://crisistextline.org>

Samaritans (UK). Call 116 123, free, 24/7. Text SHOUT to 85258. <https://samaritans.org>

9-8-8 Suicide Crisis Helpline (Canada). Call or text 9-8-8, toll-free, 24/7. English and French. <https://988.ca>

Lifeline Australia. 13 11 14. Beyond Blue: 1300 22 4636. <https://lifeline.org.au>

Find A Helpline. International crisis line directory, 70+ countries. <https://findahelpline.com>

Training

Stress First Aid for Health Care Workers. National Center for PTSD (US Department of Veterans Affairs). Free. The peer-delivered model this framework is built around. https://www.ptsd.va.gov/professional/treat/type/stress_first_aid.asp

Mental Health First Aid USA. Eight-hour certification course, delivered by licensed providers. Free in many jurisdictions. <https://mentalhealthfirstaid.org>

Practitioner Support and Instruments

ProQOL Scale (Version 5). Beth Hudnall Stamm, 2009-2012. Freely available. Validated across dozens of occupational populations. Recommended for 90-day longitudinal assessment in high-exposure roles. <https://proqol.org/proqol-measure>

OVC Vicarious Trauma Toolkit. Office for Victims of Crime. Free publicly maintained toolkit defining VT and organizational readiness assessment for first responder and victim services populations.

Open Path Psychotherapy Collective. Sessions at \$40-70 with one-time \$65 lifetime membership. All 50 states. <https://openpathcollective.org>

FindTreatment.gov. Confidential federal locator for mental health and substance use treatment. Free, no account required. Useful when the question is not affordability but finding anyone at all. <https://findtreatment.gov>

SAMHSA National Helpline. 1-800-662-4357. Free, confidential, 24/7. <https://samhsa.gov/find-help/national-helpline>

PART 8 — SOURCES AND CITATIONS

Academic and peer-reviewed literature is listed first within each category, followed by institutional doctrine, then practitioner and reference resources.

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The author welcomes corrections, additions, and challenges to the inference-based role-to-injury mapping in Part 2.

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Records and Personal Data

This framework asks practitioners to notice changes in colleagues over time. It doesn't ask anyone to create or retain records about another person, and it advises against it. Information about an identified colleague's wellbeing is likely to constitute sensitive or special category personal data in many jurisdictions, and handling it may trigger obligations that this document doesn't address and can't resolve.

Any organization implementing assessments, escalation logs, case records, or any other retained documentation in connection with these practices is responsible for its own lawful basis, retention limits, access controls, data protection impact assessment where required, and compliance with employment and disability law in every jurisdiction where it operates. Obtain privacy and legal review before collecting anything.

Vignettes Are Fictionalized

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Crisis Resources and Currency

The crisis lines, helplines, and support resources in Part 7 were verified as accurate on the date of publication of this version. Contact numbers, service hours, coverage, and operating

organizations change over time and are outside the author's control. No representation is made that any listed resource will be available, reachable, or appropriate at the time of use, and readers in an emergency should use local emergency services. Readers who identify an inaccurate or discontinued resource are asked to notify the author so it can be corrected.

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Contacting the Author

I'm glad to hear from people about this work. Corrections, disagreements, adaptations, arguments about where I've got something wrong, questions about how a section is meant to be used, all of it is welcome. The address is [humanirframework \[at\] gmail \[dot\] com](mailto:humanirframework@gmail.com).

What I can't do is provide individual support. I'm not a clinician, I'm not trained to help someone in crisis, and I can't be reliably available. Messages to that address may go unanswered for long stretches, and it isn't monitored. Please don't treat it as a way to reach someone when you need help now.

If you need help now, use the crisis resources at the front of this document. They're staffed, they're free, and the people answering are trained for exactly that.

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